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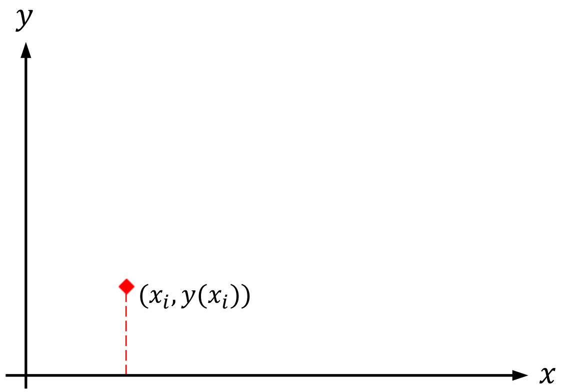

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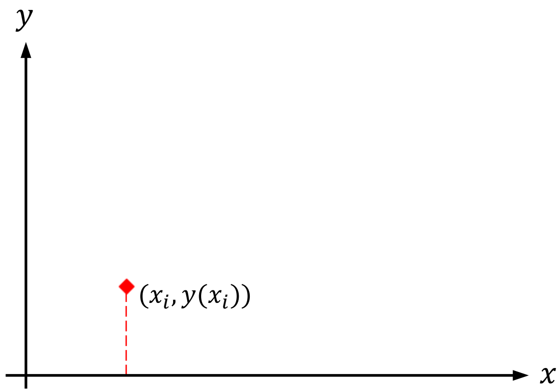

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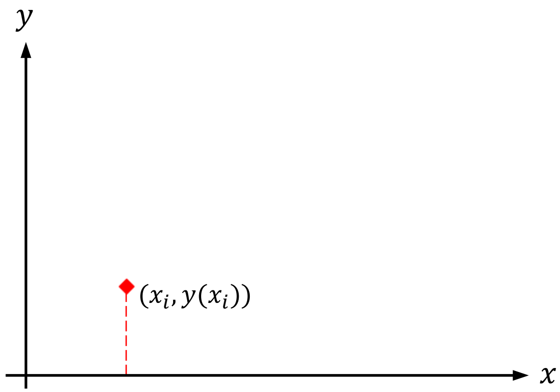

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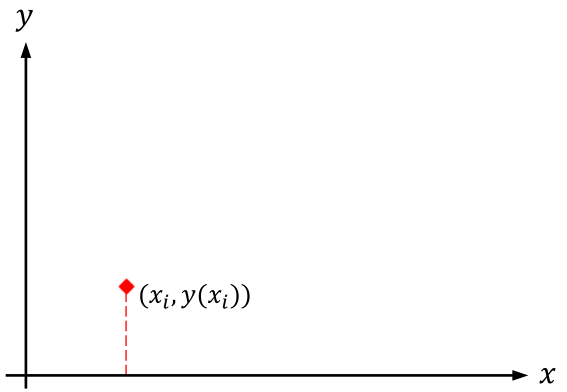
$\boxed{\text{One-Step Method}}$

$$y' = f(x, y) \;\rightarrow\; \int_{y(x_i)}^{y(x_{i+1})} dy = \int_{x_i}^{x_{i+1}} f(x, y) \cdot dx$$



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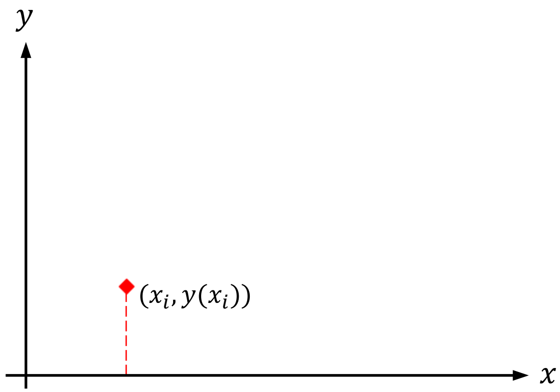
One-Step Method

$$y(x_{i+1}) = y(x_i) + \int_{x_i}^{x_{i+1}} f(x, y) \cdot dx$$



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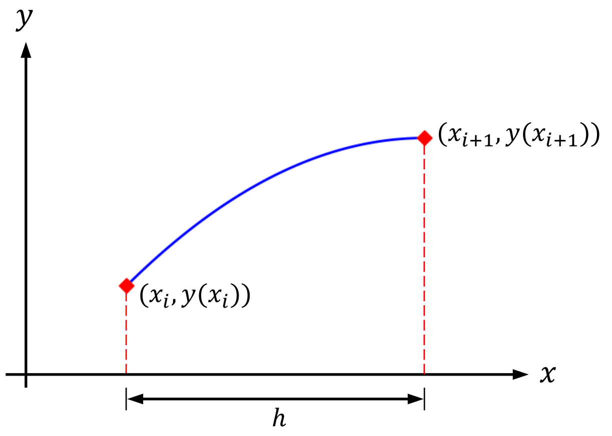

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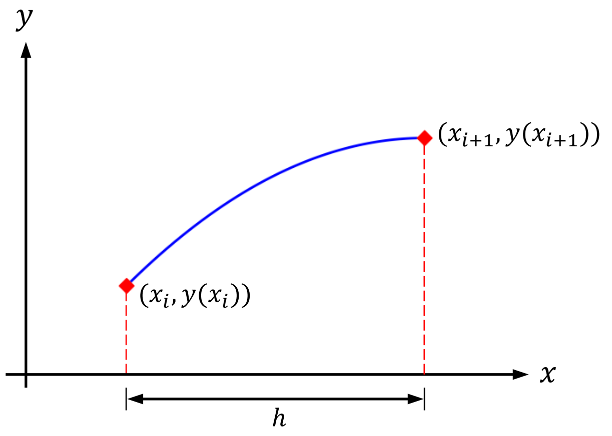

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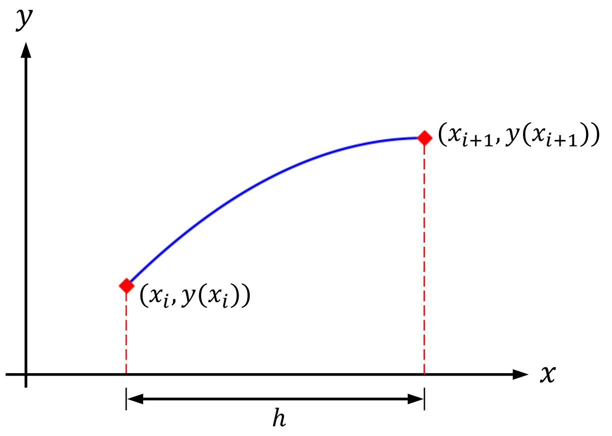
Euler Method



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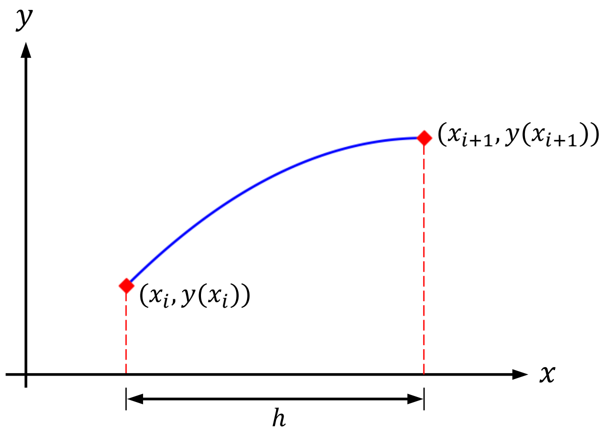

One-Step Method

$$y_{i+1} = y_i + \int_{x_i}^{x_{i+1}} f(x, y) \cdot dx$$

Euler Method

Explicit Euler Method: $y_{i+1} = y_i + h \cdot f(x_i, y_i)$

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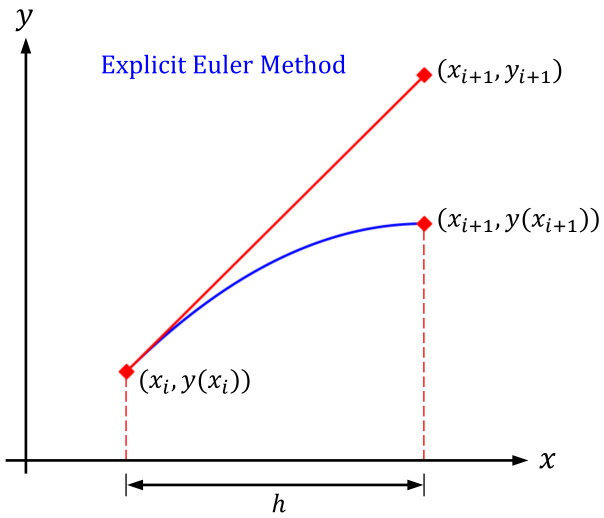

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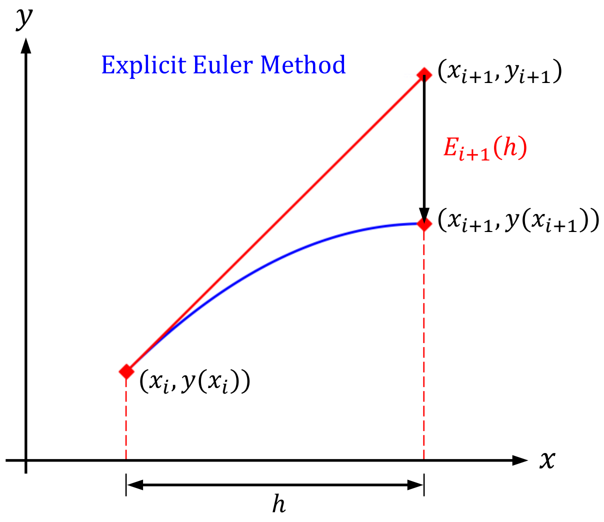

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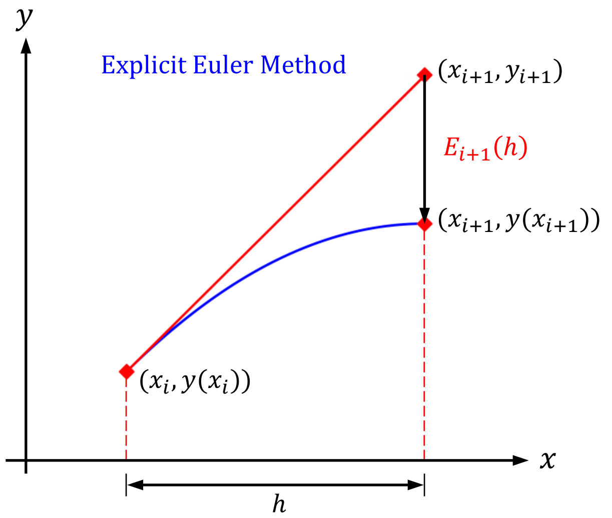

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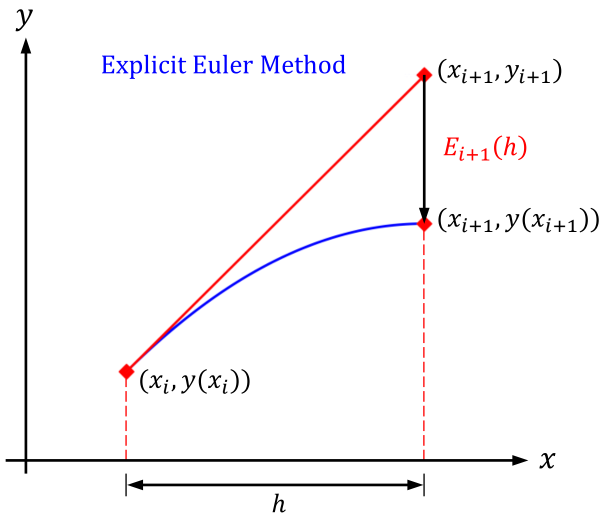

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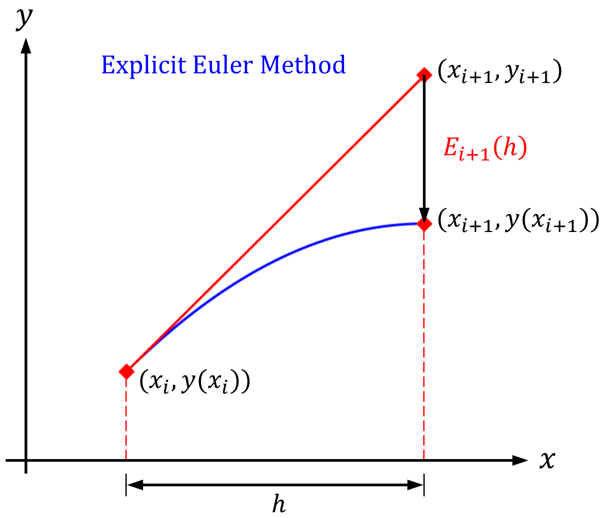

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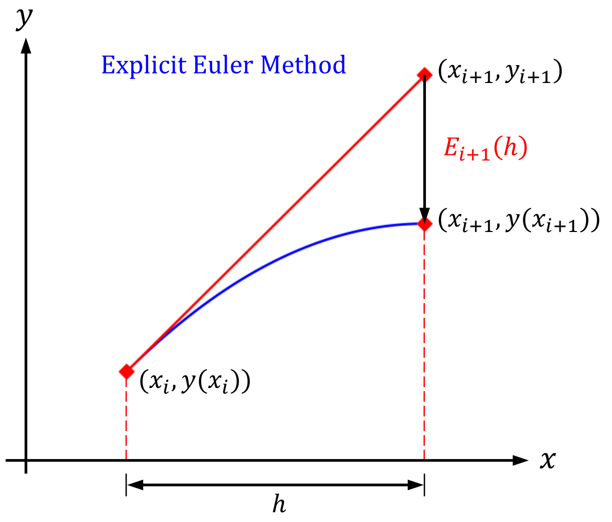

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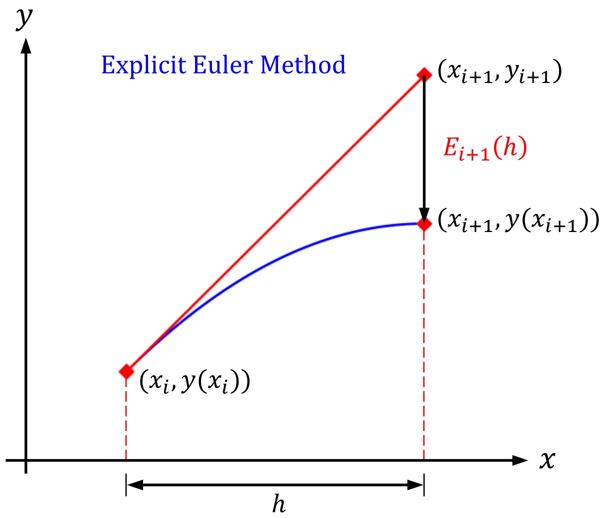

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$w = \frac{1}{2}$ → Trapezium Rule Method (Second Order)

One-Step Method

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Domain of Stability

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Initial Value Problem: $y' = \lambda y; \quad y(0) = y_0 + \varepsilon_0$

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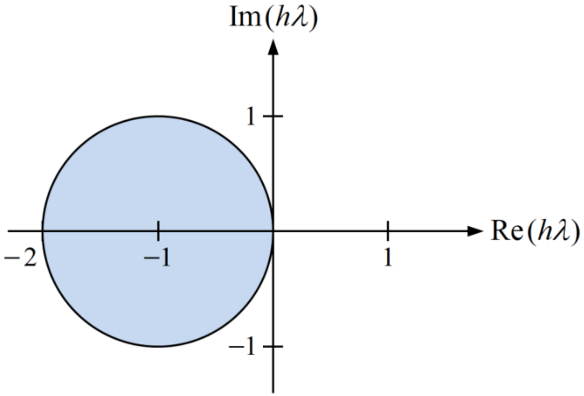

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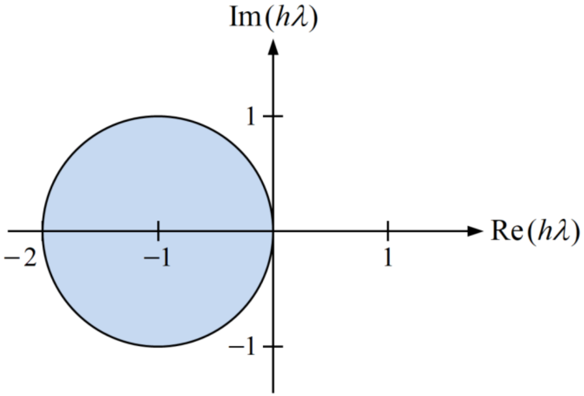

Implicit Euler Method:

$$\varepsilon_i = \frac{1}{(1 - h\lambda)^i} \cdot \varepsilon_0$$

One-Step Method

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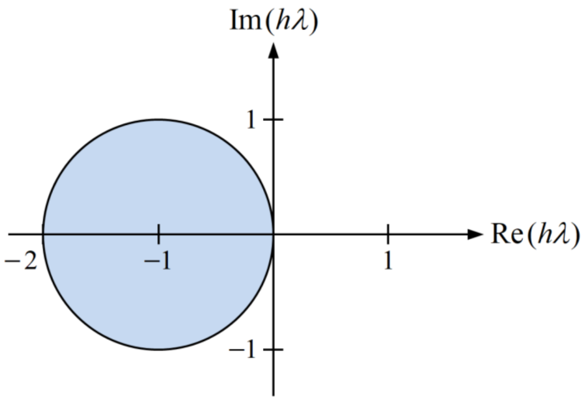

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$$\varepsilon_i = \frac{1}{(1 - h\lambda)^i} \cdot \varepsilon_0 \quad \rightarrow \quad \left|\frac{1}{1 - h\lambda}\right| \leq 1$$

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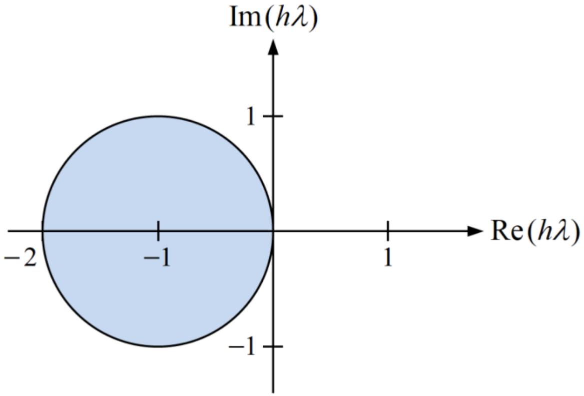

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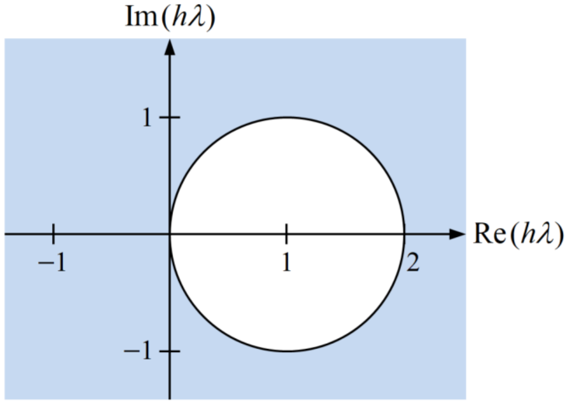

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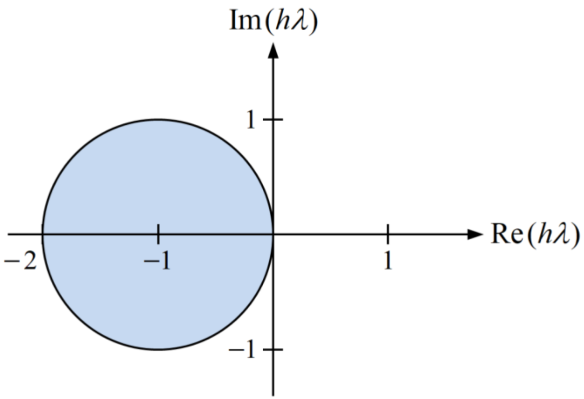

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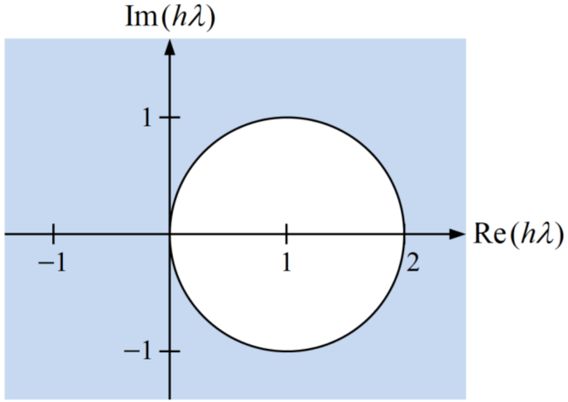

Trapezium Rule Method:

$$\varepsilon_i = \left(\frac{1 + \frac{h\lambda}{2}}{1 - \frac{h\lambda}{2}}\right)^i \cdot \varepsilon_0$$

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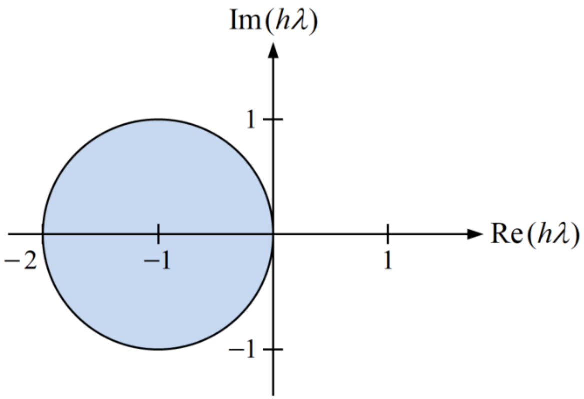

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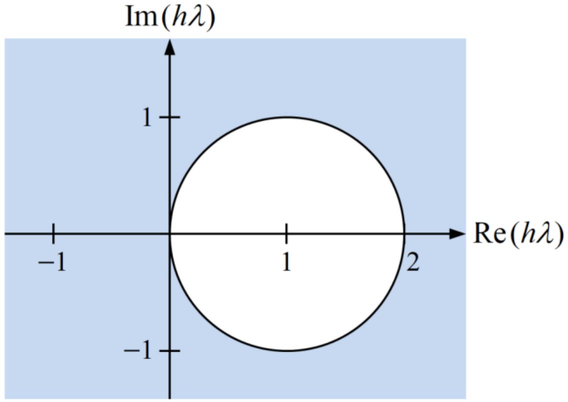

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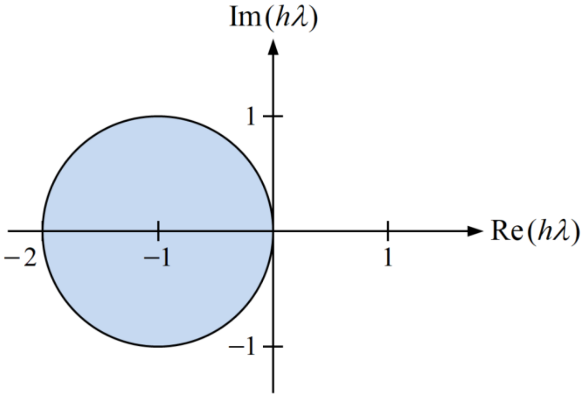

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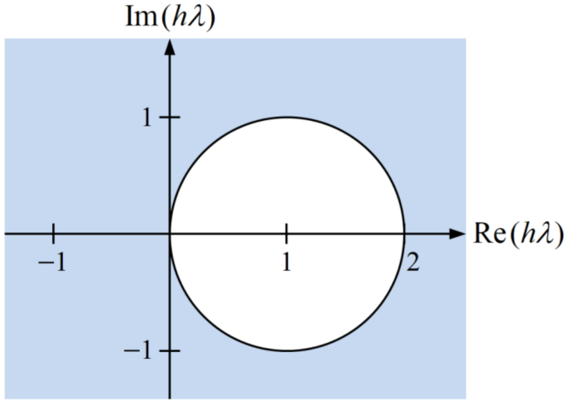

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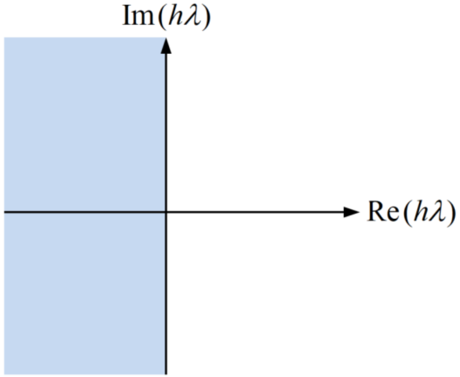

One-Step Method

Example 07-02: Approximate the numerical solution of the following initial value problem using explicit Euler method with $\alpha = 2,20$ for $0 \le x \le 1$ with step size $h = 0.1$.

$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \le x \le 1$$

One-Step Method

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Exact Solution: $y(x) = e^{-\alpha \cdot x} + x^2$

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$$f(x, y) = -\alpha \cdot (y - x^2) + 2x, N = \frac{1 - 0}{0.1} = 10, x_0 = 0, y_0 = 1$$

One-Step Method

Example 07-02: Approximate the numerical solution of the following initial value problem using explicit Euler method with $\alpha = 2{,}20$ for $0 \leq x \leq 1$ with step size $h = 0.1$.

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One-Step Method

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One-Step Method

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$$y_2 = y_1 + h \cdot [-2 \cdot (y_1 - x_1^2) + 2x_1]$$

One-Step Method

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One-Step Method

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One-Step Method

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One-Step Method

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$$\begin{aligned} y_2 &= y_1 + h \cdot [-20 \cdot (y_1 - x_1^2) + 2x_1] \\ &= -1 + 0.1 \cdot [-20 \cdot (-1 - 0.1^2) + 2 \cdot (0.1)] \end{aligned}$$

One-Step Method

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One-Step Method

Example 07-03: Approximate the numerical solution of the following initial value problem using implicit Euler method with $\alpha = 2{,}20$ for $0 \leq x \leq 1$ with step size $h = 0.1$.

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One-Step Method

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One-Step Method

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(Linear Equation)

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$$(1 + h\alpha) \cdot y_{i+1} = y_i + h \cdot (\alpha x_{i+1}^2 + 2x_{i+1})$$

One-Step Method

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$$y_{i+1} = \frac{y_i + h \cdot (\alpha x_{i+1}^2 + 2x_{i+1})}{1 + h\alpha}$$

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(Linear Equation)

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$$y_1 = \frac{y_0 + h \cdot (\alpha x_1^2 + 2x_1)}{1 + h\alpha}$$

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$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \leq x \leq 1$$

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$$f(x, y) = -\alpha \cdot (y - x^2) + 2x, N = \frac{1 - 0}{0.1} = 10, x_0 = 0, y_0 = 1$$

Implicit Euler Method:

(Linear Equation)

$$y_{i+1} = y_i + h \cdot f(x_{i+1}, y_{i+1})$$

$$y_{i+1} = y_i + h \cdot \left[-\alpha \cdot (y_{i+1} - x_{i+1}^2) + 2x_{i+1}\right]$$

$$(1 + h\alpha) \cdot y_{i+1} = y_i + h \cdot (\alpha x_{i+1}^2 + 2x_{i+1})$$

$$y_{i+1} = \frac{y_i + h \cdot (\alpha x_{i+1}^2 + 2x_{i+1})}{1 + h\alpha}$$

$$y_1 = \frac{y_0 + h \cdot (\alpha x_1^2 + 2x_1)}{1 + h\alpha}$$

$$y_2 = \frac{y_1 + h \cdot (\alpha x_2^2 + 2x_2)}{1 + h\alpha}$$

One-Step Method

<u>Example 07-03</u>: Approximate the numerical solution of the following initial value problem using implicit Euler method with $\alpha = 2{,}20$ for $0 \le x \le 1$ with step size $h = 0.1$.

$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \le x \le 1$$

Exact Solution: $y(x) = e^{-\alpha \cdot x} + x^2$

$$f(x,y) = -\alpha \cdot (y - x^2) + 2x, N = \frac{1-0}{0.1} = 10, x_0 = 0, y_0 = 1$$

Implicit Euler Method:

(Linear Equation)

$$y_{i+1} = y_i + h \cdot f(x_{i+1}, y_{i+1})$$

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$$y_{i+1} = \frac{y_i + h \cdot (\alpha x_{i+1}^2 + 2x_{i+1})}{1 + h\alpha}$$

$$y_1 = \frac{y_0 + h \cdot (\alpha x_1^2 + 2x_1)}{1 + h\alpha}$$

$$y_2 = \frac{y_1 + h \cdot (\alpha x_2^2 + 2x_2)}{1 + h\alpha}$$

$$y_3 = \frac{y_2 + h \cdot (\alpha x_3^2 + 2x_3)}{1 + h\alpha}$$

$$\vdots$$

One-Step Method

<u>Example 07-04</u>: Approximate the numerical solution of the following initial value problem using trapezium rule method with $\alpha = 2,20$ for $0 \le x \le 1$ with step size $h = 0.1$.

$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \le x \le 1$$

One-Step Method

Example 07-04: Approximate the numerical solution of the following initial value problem using trapezium rule method with $\alpha = 2{,}20$ for $0 \leq x \leq 1$ with step size $h = 0.1$.

$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \leq x \leq 1$$

Exact Solution: $y(x) = e^{-\alpha \cdot x} + x^2$

One-Step Method

Example 07-04: Approximate the numerical solution of the following initial value problem using trapezium rule method with $\alpha = 2{,}20$ for $0 \le x \le 1$ with step size $h = 0.1$.

$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \le x \le 1$$

Exact Solution: $y(x) = e^{-\alpha \cdot x} + x^2$

$$f(x, y) = -\alpha \cdot (y - x^2) + 2x, N = \frac{1 - 0}{0.1} = 10, x_0 = 0, y_0 = 1$$

One-Step Method

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$$y' = -\alpha \cdot (y - x^2) + 2x, y(0) = 1; \quad 0 \le x \le 1$$

Exact Solution: $y(x) = e^{-\alpha \cdot x} + x^2$

$$f(x, y) = -\alpha \cdot (y - x^2) + 2x, N = \frac{1 - 0}{0.1} = 10, x_0 = 0, y_0 = 1$$

Trapezium Rule Method: $y_{i+1} = y_i + h \cdot \left[\frac{1}{2} \cdot f(x_i, y_i) + \frac{1}{2} \cdot f(x_{i+1}, y_{i+1})\right]$

One-Step Method

Example 07-04: Approximate the numerical solution of the following initial value problem using trapezium rule method with $\alpha = 2{,}20$ for $0 \le x \le 1$ with step size $h = 0.1$.

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$$y_{i+1} = y_i + h \cdot \left[\tfrac{1}{2} \cdot f(x_i, y_i) + \tfrac{1}{2} \cdot f(x_{i+1}, y_{i+1})\right]$$

$$y_{i+1} = y_i + \frac{h}{2} \cdot \left[-\alpha \cdot (y_i - x_i^2) + 2x_i\right] + \frac{h}{2} \cdot \left[-\alpha \cdot (y_{i+1} - x_{i+1}^2) + 2x_{i+1}\right]$$

One-Step Method

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(Linear Equation)

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One-Step Method

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$$y_{i+1} = y_i + h \cdot \left[\tfrac{1}{2} \cdot f(x_i, y_i) + \tfrac{1}{2} \cdot f(x_{i+1}, y_{i+1})\right]$$

(Linear Equation)

$$y_{i+1} = y_i + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_i - x_i^2) + 2x_i\right] + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_{i+1} - x_{i+1}^2) + 2x_{i+1}\right]$$

$$\left(1 + \tfrac{h\alpha}{2}\right) \cdot y_{i+1} = y_i + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_i - x_i^2) + 2x_i\right] + \tfrac{h}{2} \cdot (\alpha x_{i+1}^2 + 2x_{i+1})$$

One-Step Method

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Trapezium Rule Method:

(Linear Equation)

$$y_{i+1} = y_i + h \cdot \left[\tfrac{1}{2} \cdot f(x_i, y_i) + \tfrac{1}{2} \cdot f(x_{i+1}, y_{i+1})\right]$$

$$y_{i+1} = y_i + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_i - x_i^2) + 2x_i\right] + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_{i+1} - x_{i+1}^2) + 2x_{i+1}\right]$$

$$\left(1 + \tfrac{h\alpha}{2}\right) \cdot y_{i+1} = y_i + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_i - x_i^2) + 2x_i\right] + \tfrac{h}{2} \cdot (\alpha x_{i+1}^2 + 2x_{i+1})$$

$$y_{i+1} = \frac{y_i + \tfrac{h}{2} \cdot \left[-\alpha \cdot (y_i - x_i^2) + 2x_i\right] + \tfrac{h}{2} \cdot (\alpha x_{i+1}^2 + 2x_{i+1})}{1 + \tfrac{h\alpha}{2}}$$